	GLOBAL ECONOMICS **GRADE: 11TH** **LEARNING EVIDENCE 1** **CONFIDENCE INTERVAL PLANNING** **TEACHER'S NAME: Nicolás López Cuéllar**	**SECOND TERM 2025–2026**

Learning objective: Apply statistical inference to estimate population parameters by constructing, interpreting, and justifying confidence intervals and required sample sizes based on sample data.	**Assessment criteria:** C1: Compute the sample mean and sample standard deviation. C2: Distinguish between sample statistics and population parameters. C3: Explain why interval estimation is preferred over a single point estimate. C4: Interpret the meaning of a X% confidence interval in context. C5: Construct a X% confidence interval using known population standard deviation. C6: Establish the sample size so the error will not exceed a specified amount. C7: Construct the confidence interval for a proportion.

Criteria assessment

In this hard version, problems 1–4 assess criteria C1, C5, and C6; problems 5–8 assess criteria C2, C3, and C4; and problems 9–12 assess criterion C7. A criterion is considered passed when it is correctly activated in at least four problems.

1. (C1, C5, C6) Maintenance Cost Comparison Across Two Rail Depots

A regional rail authority compares average monthly maintenance cost per trainset (thousand USD) in two depots after introducing predictive maintenance software. The grouped data report maintenance-cost values together with their frequencies, where each cost value occurs in a certain number of trainsets.

Depot A: The data correspond to 27 trainsets. Each maintenance-cost value below is measured in thousand USD, and the stated frequency indicates how many trainsets had that value. Cost = 72 (thousand USD) occurs in 5 trainsets; cost = 78 (thousand USD) occurs in 7 trainsets; cost = 85 (thousand USD) occurs in 6 trainsets; cost = 92 (thousand USD) occurs in 5 trainsets; and cost = 100 (thousand USD) occurs in 4 trainsets.

Depot B: The data correspond to 28 trainsets. Each maintenance-cost value below is measured in thousand USD, and the stated frequency indicates how many trainsets had that value. Cost = 68 (thousand USD) occurs in 4 trainsets; cost = 75 (thousand USD) occurs in 6 trainsets; cost = 83 (thousand USD) occurs in 8 trainsets; cost = 90 (thousand USD) occurs in 6 trainsets; and cost = 98 (thousand USD) occurs in 4 trainsets.

From long-run accounting records, the population standard deviation is known to be $\sigma = 14$ thousand USD for both depots.

1. Calculate the sample mean and sample standard deviation for each depot using the provided grouped data (C1).
2. Construct 90%, 95%, and 99% confidence intervals for each depot mean using known σ, and report the margin of error for each interval (C5).
3. Compute required sample sizes for target margins of error $E = 3.5$ and $E = 2.8$ at 90%, 95%, and 99%, and determine additional observations needed for each depot (C6).

2. (C1, C5, C6) Fraud-Review Time Comparison by Team

A digital payments platform tracks average fraud-review handling time (minutes) for two independent teams. The grouped data are reported using class intervals, and the frequency for each interval indicates how many cases fall within that time range.

Team A: The data correspond to 38 cases. Each interval below is measured in minutes, and the stated frequency is the number of cases in that interval. 42–46 minutes corresponds to 6 cases; 47–51 minutes corresponds to 9 cases; 52–56 minutes corresponds to 10 cases; 57–61 minutes corresponds to 8 cases; and 62–66 minutes corresponds to 5 cases.

Team B: The data correspond to 40 cases. Each interval below is measured in minutes, and the stated frequency is the number of cases in that interval. 40–44 minutes corresponds to 5 cases; 45–49 minutes corresponds to 8 cases; 50–54 minutes corresponds to 11 cases; 55–59 minutes corresponds to 9 cases; and 60–64 minutes corresponds to 7 cases.

Assume known population standard deviation $\sigma = 6.5$ minutes.

1. Calculate the sample mean and sample standard deviation for each team using the provided grouped data (C1).

2. Construct 90%, 95%, and 99% confidence intervals for each team mean using known σ, and report the margin of error for each interval (C5).

3. Compute required sample sizes for $E = 1.2$ and $E = 0.9$ at 90%, 95%, and 99%, and determine additional cases needed for each team (C6).

3. (C1, C5, C6) Dispatch Completion Time in an Expansion Zone

An emergency logistics center records warehouse dispatch completion times (hours) for one expansion zone. A random sample of 18 dispatches is: 31, 29, 34, 28, 33, 35, 30, 32, 36, 27, 34, 31, 29, 33, 30, 37, 28, 32. Known population standard deviation is $\sigma = 5.8$ hours.

1. Calculate the sample mean and sample standard deviation for the dispatch-time sample (C1).

2. Construct 90%, 95%, and 99% confidence intervals for the population mean completion time using known σ, and report the margin of error for each interval (C5).

3. Compute required sample sizes for target errors $E = 2.0$ and $E = 1.5$ at 90%, 95%, and 99%, and determine additional dispatches needed (C6).

4. (C1, C5, C6) Audit Value Recovery Comparison Across Field Units

A tax administration compares average daily audit value recovered (thousand USD) by two independent field units.

Unit A (15 days): 112, 108, 115, 121, 117, 109, 114, 118, 116, 113, 120, 111, 119, 107, 122.

Unit B (18 days): 105, 109, 111, 114, 108, 112, 116, 110, 113, 107, 115, 118, 106, 109, 117, 111, 114, 108.

Assume known $\sigma = 9.2$ thousand USD in both units.

1. Calculate the sample mean and sample standard deviation for each unit using the provided daily values (C1).

2. Construct 90%, 95%, and 99% confidence intervals for each unit mean using known σ, and report the margin of error for each interval (C5).

3. Compute required sample sizes for $E = 3.0$ and $E = 2.2$ at 90%, 95%, and 99%, and determine additional observations needed for each unit (C6).

5. (C2, C3, C4) Fintech Repayment Delay Supervision Analysis

A central bank supervision office evaluates a fintech lender. One team reports from a recent compliance review: average repayment delay was 4.7 days with spread 2.1 days, and a 95% confidence interval for the mean delay was [4.1, 5.3]. A second team cites long-run portfolio values from the full national borrower base: true average delay = 4.9 days and true standard deviation = 2.4 days. The policy division also references a strategic threshold: "acceptable mean delay below 5.0 days."

Explicit analysis tasks: 1) Classify every numerical quantity above as a statistic from observed data, a fixed characteristic of the full borrower base, or a policy benchmark, and justify (C2). 2) Using the reported interval for mean repayment delay, answer all of the following (C3): (i) state the point estimate for the mean repayment delay; (ii) explain, in this supervision context, why reporting only that point estimate is insufficient compared to reporting the interval; (iii) compute the interval width and explain what it says about precision; (iv) propose two concrete changes to the data-collection or measurement design that would narrow the interval, and justify briefly. 3) State which mean values are ruled out/plausible under [4.1, 5.3] and explain "95% confidence" in repeated-trial language (C4).

6. (C2, C3, C4) Border Port Turnaround Analysis for Policy Decisions

A national logistics ministry monitors truck turnaround time at border ports. A pilot review from Port North gives mean 11.8 hours and 90% CI [10.9, 12.7]. A separate audit from Port South gives mean 12.4 hours and 90% CI [11.6, 13.2]. Historic all-port values for the old regime were true average turnaround = 12.1 hours and true standard deviation = 3.0 hours. A policy KPI requires maintaining average turnaround at or below 12.0 hours.

Explicit analysis tasks: 1) Classify observed-data quantities, full-system characteristics, and policy quantities (C2). 2) Using the reported intervals for mean turnaround time, answer all of the following (C3): (i) state the point estimate for Port North and the point estimate for Port South; (ii) explain, in this policy-decision context, why reporting only those point estimates is insufficient compared to reporting intervals; (iii) compare precision by computing the widths of the North and South intervals and identify which report is more precise; (iv) propose two concrete changes to the data-collection or measurement design that would narrow the interval(s), and justify briefly. 3) For each port, state ruled-out/plausible means and explain operational meaning of 90% confidence (C4).

7. (C2, C3, C4) Social Protection Processing-Time Channel Comparison

A social protection agency studies benefit-payment processing time across two digital channels. Channel A monitoring report gives mean 6.3 days with CI [5.8, 6.8] at 95%. Channel B audit report gives mean 6.0 days with CI [5.2, 6.8] at 95%. Legacy national values before automation were true average processing time = 6.7 days and true standard deviation = 1.9 days. The ministry also publishes two benchmarks: warning level 6.5 days, excellent level 6.0 days.

Explicit analysis tasks: 1) Categorize all values into observed-data statistics, fixed national characteristics, and policy benchmarks (C2). 2) Using the reported intervals for mean processing time, answer all of the following (C3): (i) state the point estimate for Channel A and the point estimate for Channel B; (ii) explain, in this social-protection context, why reporting only those point estimates is insufficient compared to reporting intervals; (iii) compare precision by computing the widths of the Channel A and Channel B intervals and identify which report is more precise; (iv) propose two concrete changes to the data-collection or measurement design that would narrow the interval(s), and justify briefly. 3) Determine plausible/ruled-out means per channel and explain repeated-trial meaning of 95% confidence (C4).

8. (C2, C3, C4) External Fund Manager Return Reporting Analysis

A sovereign wealth monitoring office evaluates monthly return volatility reporting from two external fund managers. Manager Alpha reports a mean monthly return of 1.4% with 95% CI [0.8%, 2.0%]. Manager Beta reports a mean of 1.1% with 95% CI [0.2%, 2.0%]. Historical long-run values from the full investment universe are true average monthly return = 1.3% and true standard deviation = 3.5%. The board uses two policy thresholds: "underperforming" if mean return $< 1.0\%$ and "stretch target" if mean return $\geq 1.8\%$.

Explicit analysis tasks: 1) Classify all quantities into observed-data statistics, fixed universe characteristics, and policy thresholds (C2). 2) Using the reported intervals for mean monthly return, answer all of the following (C3): (i) state the point estimate for Manager Alpha and the point estimate for Manager Beta; (ii) explain, in this strategic-allocation context, why reporting only those point estimates is insufficient compared to reporting intervals; (iii) compare precision by computing the widths of the Alpha and Beta intervals and identify which report is more precise; (iv) propose two concrete changes to the data-collection or measurement design that would narrow the interval(s), and justify briefly. 3) For each manager, state ruled-out and plausible values and explain repeated-trial meaning of 95% confidence (C4).

9. (C7) Supplier On-Time Submission Proportion by Region

A public procurement platform reports supplier on-time document submission rates for two regions. Region East sample: 420 suppliers with 246 on-time. Region West sample: 380 suppliers with 186 on-time.

1. Construct 90%, 95%, and 99% confidence intervals for each region's population proportion, and report the margin of error for each interval (C7).

10. (C7) Digital Tax Filing Completion by Business Subgroup

A municipal e-governance program evaluates digital tax filing completion by business size subgroup in one city. Small firms: $n = 250$, completed $x = 142$. Medium firms: $n = 200$, completed $x = 96$. Large firms: $n = 150$, completed $x = 57$.

1. Construct 90% and 95% confidence intervals for each subgroup proportion, and report the margin of error for each interval (C7).

11. (C7) Defective Shipment Proportion Across Two Quarterly Audits

A pharmaceutical procurement unit tracks defective shipment proportion in two independent quarterly audits. Quarter 1: $n_1 = 520$, defectives $x_1 = 57$. Quarter 2: $n_2 = 610$, defectives $x_2 = 52$.

1. Construct 95% and 99% confidence intervals for each quarter proportion, and report the margin of error for each interval (C7).

12. (C7) Renewable Energy Adoption Proportion Across Two Districts

A national energy agency studies the proportion of households that adopted rooftop solar systems in two districts during the same year. District North: $n_1 = 480$, adopters $x_1 = 168$. District South: $n_2 = 540$, adopters $x_2 = 221$.

1. Construct 90%, 95%, and 99% confidence intervals for each district proportion, and report the margin of error for each interval (C7).